

The van der Corput property for sums of two squares

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Van der Corput sets

A sequence $\{s_n\}_{n \geq 1} \subseteq \mathbb{T}$ is *equidistributed* on the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ if

$$\lim_{N \rightarrow \infty} \frac{|\{n \leq N : s_n \in I\}|}{N} = |I|$$

for every arc $I \subseteq \mathbb{T}$.

A theorem of van der Corput states that $\{s_n\}_{n \geq 1} \subseteq \mathbb{T}$ is equidistributed if the differenced sequence $\{s_{n+h} - s_n\}_{n \geq 1}$ is equidistributed for every $h \in \mathbb{N}$.

Definition (van der Corput set)

A set $\mathcal{H} \subseteq \mathbb{N}$ is called a *van der Corput set* if a given sequence $\{s_n\}_{n \geq 1} \subseteq \mathbb{T}$ is equidistributed whenever $\{s_{n+h} - s_n\}_{n \geq 1}$ is equidistributed for every $h \in \mathcal{H}$.

Examples of van der Corput sets

- ① Integer multiples: $q\mathbb{N}$, where q is any positive integer.
 - ② Infinite difference sets: $(I - I)_+ := \{n - m : m, n \in I, n > m\}$, where $I \subseteq \mathbb{N}$ is infinite.
 - ③ Thick sets: sets containing arbitrarily long strings of consecutive integers.
 - ④ k th powers: $\{n^k : n \in \mathbb{N}\}$, where $k \in \mathbb{N}$.
 - ⑤ Intersective polynomial values: $\{|P(n)| : n \in \mathbb{N}\}$, where P is a non-constant polynomial with integer coefficients such that $P(x) \equiv 0 \pmod{q}$ has a root for every $q \in \mathbb{N}$.
 - ⑥ Shifted primes: $\{p + a : p \text{ is prime}\}$, where $a = \pm 1$.
 - ⑦ Special Beatty sequences: $\{\lfloor an^b \rfloor : n \in \mathbb{N}\}$, where $a \neq 0$ and b is irrational.
- ① was observed by Delange, ② – ⑥ are due to Kamae and Mendès-France (1978), and ⑦ is due to Boshernitzan, Kolesnik, Quas, and Wierdl (2005).

The van der Corput property

Given a set $\mathcal{H} \subseteq \mathbb{N}$, let $\mathcal{T}_{\mathcal{H}}$ denote the collection of cosine polynomials with real coefficients of the form

$$T(x) = a_0 + \operatorname{Re} \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h e(hx) = a_0 + \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h \cos(2\pi hx), \quad e(hx) := e^{2\pi i hx},$$

with the properties that $T(x) \geq 0$ for all $x \in \mathbb{R}$ and that $T(0) = 1$, and define

$$\delta_{\mathcal{H}} := \inf_{T \in \mathcal{T}_{\mathcal{H}}} a_0 = \inf_{T \in \mathcal{T}_{\mathcal{H}}} \int_0^1 T(x) dx \geq 0.$$

We say that $\mathcal{H}$ possesses the *van der Corput property* if $\delta_{\mathcal{H}} = 0$.

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We say that $\mathcal{H}$ possesses the *van der Corput property* if $\delta_{\mathcal{H}} = 0$.

Kamae and Mendès-France (1978) showed that $\mathcal{H}$ is a van der Corput set if and only if it possesses the van der Corput property.

Intersective sets

Definition (intersective set)

A set $\mathcal{H} \subseteq \mathbb{N}$ is called *intersective* if every set $A \subseteq \mathbb{N}$ satisfying $(A - A) \cap \mathcal{H} = \emptyset$ has natural density 0, i.e.,

$$A(N) := |A \cap [1, N]| = o(N),$$

where $A - A := \{a - a' : a, a' \in A\}$.

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Observation

If $\mathcal{H} \subseteq \mathbb{N}$ is intersective, then every $q \in \mathbb{N}$ divides infinitely many elements of $\mathcal{H}$.

Proof. Assume that some $q \in \mathbb{N}$ only divides finitely many elements of $\mathcal{H}$. Let m be a multiple of q greater than every multiple of q in $\mathcal{H}$. Then the set $A = m\mathbb{N}$ satisfies $(A - A) \cap \mathcal{H} = \emptyset$ while having natural density $1/m$.

Lovász's conjecture on squares

Lovász conjectured that the set of squares $\mathcal{H} = \{n^2 : n \in \mathbb{N}\}$ is intersective. This was settled independently by Furstenberg (1977) using ergodic theory and Sárközy (1978) using the Hardy–Littlewood circle method plus the density increment argument pioneered by Roth in his work on sets lacking three-term arithmetic progressions.

In fact, Sárközy proved the following quantitative result.

Theorem 2.1 (Sárközy's theorem for squares)

Let $\mathcal{H} = \{n^2 : n \in \mathbb{N}\}$. If $A \subseteq \mathbb{N}$ satisfies $(A - A) \cap \mathcal{H} = \emptyset$, then

$$A(N) \ll N(\log \log N)^{2/3}(\log N)^{-1/3}.$$

In the opposite direction, Rusza (1984) showed that there is $A \subseteq \mathbb{N}$ with

$$A(N) \gg N^c, \quad c = \frac{1}{2} \left(1 + \frac{\log 7}{\log 65} \right) = 0.733077 \dots$$

Improvements

Sárközy's bound has been improved a few times:

- ① Pintz, Steiger, and Szemerédi (1988):

$$A(N) \ll N(\log N)^{-\frac{1}{12} \log \log \log \log N}.$$

- ② Bloom and Maynard (2022):

$$A(N) \ll N(\log N)^{-c_1 \log \log \log N}, \quad c_1 > 0.$$

- ③ Green and Sawhney (2024):

$$A(N) \ll N \exp\left(-c_2(\log N)^{1/4}(\log \log N)^{-1/2}\right), \quad c_2 > 0.$$

- ④ Green and Sawhney (2025):

$$A(N) \ll N \exp\left(-c_3(\log N)^{1/2}\right), \quad c_3 > 0.$$

Analogues for intersective polynomials

Analogous results have been established for general intersective polynomial values $\mathcal{H} = \{|P(n)| : n \in \mathbb{N}\}$ with $k = \deg P \geq 2$:

① Lucier (2006):

$$A(N) \ll_P N(\log \log N)^{\rho_k/(k-1)} (\log N)^{-1/(k-1)}, \quad \rho_k = 2 + \mathbf{1}_{k=2}.$$

② Rice (2019):

$$A(N) \ll_P N(\log N)^{-c_4 \log \log \log \log N}, \quad c_4 = c_4(k) > 0.$$

③ Arala (2023):

$$A(N) \ll_P N(\log N)^{-c_5 \log \log \log N}, \quad c_5 = c_5(k) > 0.$$

④ Adajar, Agrawal, Choudhuri, Chuah, F., Hegde, Lott, Nandakumar, and Ponagandla (2026):

$$A(N) \ll_{P,\epsilon} N \exp(-c_6 (\log N)^{1/2-\epsilon}), \quad c_6 = c_6(P, \epsilon) > 0.$$

A conjecture of Erdős on shifted primes

Sárközy (1978) also showed that the set of shifted primes

$$\mathcal{H} = \{p - 1 : p \text{ is prime}\}$$

is intersective, answering a question of Erdős.

Theorem 2.2 (Sárközy's theorem for shifted primes)

Let $\mathcal{H} = \{p - 1 : p \text{ is prime}\}$. If $A \subseteq \mathbb{N}$ satisfies $(A - A) \cap \mathcal{H} = \emptyset$, then

$$A(N) \ll N(\log \log \log N)^3 (\log \log \log \log N) (\log \log N)^{-2}.$$

On the other hand, Rusza (1984) showed that there is $A \subseteq \mathbb{N}$ with

$$A(N) \gg \exp \left((\log 2 - o(1)) \frac{\log N}{\log \log N} \right).$$

Road to power saving

- ① Lucier (2008):

$$A(N) \ll N \left(\frac{(\log \log \log N)^3 (\log \log \log \log N)}{\log \log N} \right)^{\log \log \log \log \log N}.$$

- ② Ruzsa and Sanders (2008):

$$A(N) \ll N \exp(-c_7 (\log N)^{1/4}), \quad c_7 > 0.$$

- ③ Wang (2020):

$$A(N) \ll N \exp(-c_8 (\log N)^{1/3}), \quad c_8 > 0.$$

- ④ Green (2023):

$$A(N) \ll N^{1-c_9}, \quad c_9 > 0.$$

Thorner and Zaman (2024): $c_9 = 10^{-18}$ is admissible.

Sárközy meets van der Corput

Kamae and Mendès-France (1978) proved that given $\mathcal{H} \subseteq \mathbb{N}$,

$$A(N) \leq (\delta_{\mathcal{H}} + o(1))N$$

for every $A \subseteq \mathbb{N}$ such that $(A - A) \cap \mathcal{H} = \emptyset$. In particular, van der Corput sets are intersective!

Green achieved his power-saving bound by proving a strong quantitative version of the van der Corput property for shifted primes. Weaker results had been previously obtained by Slijepčević (2010, 2013) for both squares and shifted primes.

Sárközy via the quantitative van der Corput property

Proposition 2.3 (Sárközy via quantitative van der Corput)

Let $\mathcal{H} \subseteq \mathbb{N}$, $H \in \mathbb{Z}_{\geq 0}$, $N \in \mathbb{N}$, and

$$T_H(x) = a_0 + \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h \cos(2\pi hx) \in \mathcal{T}_{\mathcal{H}}.$$

Then $A(N) \leq (N + H)a_0$ for every $A \subseteq \mathbb{N}$ satisfying $(A - A) \cap \mathcal{H} = \emptyset$.

In practice, we take $H = N$ and seek cosine polynomials $T_N(x) \in \mathcal{T}_{\mathcal{H}}$ with $a_0 = a_0(N)$ decaying as fast as possible.

Sárközy via the quantitative van der Corput property

Proof. Let $M = N + H$.

1

$$\overbrace{A(N)^2 T_N(0)}^{r=0 \text{ term}} \underbrace{=1} \leq R_N := \sum_{r \in \mathbb{Z}/M\mathbb{Z}} \left| \sum_{\substack{n \leq N \\ n \in A}} e\left(\frac{rn}{M}\right) \right|^2 \underbrace{T_N\left(\frac{r}{M}\right)}_{\geq 0}.$$

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2

$$\begin{aligned} R_N &= \sum_{\substack{m, n \leq N \\ m, n \in A}} \sum_{r \in \mathbb{Z}/M\mathbb{Z}} \overbrace{e\left(\frac{r(m-n)}{M}\right)}^{=M\mathbf{1}_{m \equiv n \pmod{M}}} \left(a_0 + \operatorname{Re} \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h e\left(\frac{rh}{M}\right) \right) \\ &= A(N)Ma_0 + \operatorname{Re} \sum_{\substack{m, n \leq N \\ m, n \in A}} \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h \sum_{r \in \mathbb{Z}/M\mathbb{Z}} e\left(\frac{r(m-n-h)}{M}\right). \end{aligned}$$

Sárközy via the quantitative van der Corput property

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$$A(N)^2 \leq A(N)Ma_0 + \operatorname{Re} \sum_{\substack{m,n \leq N \\ m,n \in A}} \sum_{\substack{h \leq H \\ h \in \mathcal{H}}} a_h \overbrace{\sum_{r \in \mathbb{Z}/M\mathbb{Z}} e\left(\frac{r(m-n-h)}{M}\right)}^{=M\mathbf{1}_{m-n-h \equiv 0 \pmod{M}}}.$$

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4

- $1 \leq m, n \leq N, 1 \leq h \leq H \implies |m - n - h| < N + H = M.$
- $(A - A) \cap \mathcal{H} = \emptyset \implies m - n - h \not\equiv 0 \pmod{M}.$

5

$$A(N)^2 \leq A(N)Ma_0 + 0.$$

So $A(N) \leq Ma_0 = (N + H)a_0.$

Why $\square + \square$?

Let

$$\mathcal{S} := \{d \in \mathbb{N} : d = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}\}, \quad \mathcal{S}_N := \mathcal{S} \cap [1, N].$$

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- Equidistribution among suitable reduced residue classes.
- Generalizations such as quadratic and higher-degree homogeneous forms.
- $\mathcal{S}$ is a van der Corput set (and hence an intersective set).
- Rice (2020) has studied it.

The van der Corput property for sums of two squares

Recall

$$\mathcal{S}_N = \{1 \leq d \leq N : d = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}\}.$$

Theorem 3.1 (F.–Lott, 2026)

For every $\epsilon > 0$ there is a constant $C_\epsilon > 0$ such that for every $N \in \mathbb{N}$, there are coefficients $c_d \geq 0$, supported on $\mathcal{S}_N$, satisfying

$$\sum_{d \in \mathcal{S}_N} c_d = 1$$

and

$$\operatorname{Re} \sum_{d \in \mathcal{S}_N} c_d e(d\theta) \geq -C_\epsilon N^{-1/8+\epsilon}, \quad \forall \theta \in \mathbb{T}.$$

Sárközy for sums of two squares

Corollary 3.2 (F.–Lott, 2026)

Given any $\epsilon > 0$,

$$A(N) \ll_{\epsilon} N^{7/8+\epsilon}$$

holds for every $A \subseteq \mathbb{N}$ such that $(A - A) \cap \mathcal{S} = \emptyset$.

This improves upon the quasipolynomial bound

$$A(N) \ll N \exp(-c_{10} \sqrt{\log N}), \quad c_{10} > 0,$$

due to Rice (2020).

On the other hand, Younis (2019) showed that there is some $A \subseteq \mathbb{N}$ satisfying

$$(A - A) \cap \mathcal{S} = \emptyset \quad \text{and} \quad A(N) \gg N^{1/2}.$$

(In fact, one may take $A = \{n \leq N : p \mid n\}$, where $p \equiv 3 \pmod{4}$ is a prime in $(N^{1/2}, 2N^{1/2}]$.)

Starting with a weight function

Let $H > 0$ and let $U \in C_c^\infty(\mathbb{R})$ be nonnegative with $\text{supp } U = [1, 2]$. Define

$$\mathcal{B}_H(\theta) := \frac{1}{2} \sum_{v, x, y \in \mathbb{Z}} U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) e^{-(v/H)^2} e(v^2(x^2 + y^2)\theta).$$

- $U(x/H)U(y/H)$ smoothens the sum and restricts x, y to $[H, 2H]$.
- $e^{-(v/H)^2}$ supplies convenient smooth decay that makes the contribution from large v negligible and plays a role in controlling $\arg \mathcal{B}_H(\theta)$.
- The decomposition $d = v^2(x^2 + y^2)$ deliberately amplifies the contribution from $d \in \mathcal{S}$ having nontrivial square divisors in order to control $\arg \mathcal{B}_H(\theta)$ together with $e^{-(v/H)^2}$.

Without the $v = 0$ term and the smoothing factors, $\mathcal{B}_H(\theta)$ becomes

$$\frac{1}{2} \sum_{\substack{v, x, y \in \mathbb{Z} \\ v \neq 0}} e(v^2(x^2 + y^2)\theta) = \sum_{d \geq 0} \left(\sum_{\substack{x, y \in \mathbb{Z} \\ v \geq 1}} \mathbf{1}_{v^2(x^2 + y^2) = d} \right) e(d\theta).$$

Normalizing $B_H(\theta)$

Consider the normalization

$$\mathcal{F}_H(\theta) := \frac{\mathcal{B}_H(\theta)}{\mathcal{B}_H(0)}.$$

Applying the Poisson summation formula

$$\sum_{n \in \mathbb{Z}^k} f\left(\frac{n}{H}\right) = H^k \sum_{\ell \in \mathbb{Z}^k} \widehat{f}(H\ell), \quad \forall f \in \mathcal{S}(\mathbb{R}^k),$$

gives

$$\mathcal{B}_H(0) = C_U H^3 + O_{U,A}(H^{-A})$$

for any $A > 0$, where

$$C_U := \frac{\sqrt{\pi}}{2} \int_{\mathbb{R}} U(t) dt > 0.$$

Defining the coefficients c_d

Suppose $N \in \mathbb{N}$ is large and take $H = N^{1/4}(\log N)^{-25}$. Setting

$$w_H(d) := \sum_{\substack{v \geq 1, x, y \in \mathbb{Z} \\ v^2(x^2 + y^2) = d}} U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) e^{-(v/H)^2},$$

an elementary argument shows

$$\begin{aligned} \mathcal{F}_H(\theta) &= \frac{\mathcal{B}_H(\theta)}{\mathcal{B}_H(0)} = \frac{\frac{1}{2}w_H(0) + \sum_{d \geq 1} w_H(d)e(d\theta)}{\frac{1}{2}w_H(0) + \sum_{d \geq 1} w_H(d)} \\ &\approx \frac{\sum_{d \geq 1} w_H(d)e(d\theta)}{\sum_{d \geq 1} w_H(d)} \\ &\approx \frac{\sum_{d \leq N} w_H(d)e(d\theta)}{\sum_{d \leq N} w_H(d)}. \end{aligned}$$

Defining the coefficients c_d

Define

$$c_d := \frac{w_H(d)}{\sum_{m \leq N} w_H(m)} \mathbf{1}_{d \leq N}.$$

Then $c_d \geq 0$ are supported on $\mathcal{S}_N$ and sum up to 1. It suffices to have

$$\operatorname{Re} \sum_{d \in \mathcal{S}_N} c_d e(d\theta) \geq -C_\epsilon N^{-1/8+\epsilon}, \quad \forall \theta \in \mathbb{T}.$$

Proposition 3.3

For every $\epsilon > 0$, there are constants $C_{U,\epsilon}, H_0(U,\epsilon) > 0$ such that for every $H \geq H_0(U,\epsilon)$,

$$\operatorname{Re}(\mathcal{F}_H(\theta)) = \frac{\operatorname{Re}(\mathcal{B}_H(\theta))}{\mathcal{B}_H(0)} \geq -C_{U,\epsilon} H^{-1/2+\epsilon}, \quad \forall \theta \in \mathbb{T}.$$

Estimation of $\mathcal{B}_H(\theta)$: the major arcs

Given $H \geq 1$, and $0 < \sigma < 1/4$, we define the major arcs by

$$\mathfrak{M} := \bigcup_{1 \leq q \leq Q} \bigcup_{\substack{1 \leq a \leq q \\ (a, q) = 1}} \left\{ \theta \in \mathbb{T} : \left\| \theta - \frac{a}{q} \right\| \leq \frac{Q}{qH^4} \right\}, \quad Q := H^{1-4\sigma}.$$

For each $\theta \in \mathfrak{M}$, write $\theta = a/q + \beta$, where $(a, q) = 1$ and $|\beta| \leq \frac{Q}{qH^4}$. Then

$$\begin{aligned} \mathcal{B}_H(\theta) &= \frac{1}{2} \sum_{v, x, y \in \mathbb{Z}} U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) e^{-(v/H)^2} e\left(v^2(x^2 + y^2) \left(\frac{a}{q} + \beta\right)\right) \\ &= \frac{1}{2} \sum_{v, x, y \in \mathbb{Z}} e\left(\frac{a}{q} v^2(x^2 + y^2)\right) \Phi_\beta(v, x, y) \\ &= \frac{1}{2} \sum_{r, s, t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{a}{q} r^2(s^2 + t^2)\right) \sum_{\substack{v, x, y \in \mathbb{Z} \\ v \equiv r \pmod{q} \\ x \equiv s \pmod{q} \\ y \equiv t \pmod{q}}} \Phi_\beta(v, x, y). \end{aligned}$$

Estimation of $\mathcal{B}_H(\theta)$: the major arcs

Recall

$$\mathcal{B}_H(\theta) = \frac{1}{2} \sum_{r,s,t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{a}{q}r^2(s^2 + t^2)\right) \sum_{\substack{v,x,y \in \mathbb{Z} \\ v \equiv r \pmod{q} \\ x \equiv s \pmod{q} \\ y \equiv t \pmod{q}}} \Phi_\beta(v, x, y),$$

where

$$\Phi_\beta(v, x, y) := U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) \exp\left(-\frac{v^2}{H^2} + 2\pi i \beta v^2(x^2 + y^2)\right) \in \mathcal{S}(\mathbb{R}^3).$$

Applying Poisson summation yields

$$\mathcal{B}_H(\theta) = \frac{1}{2} q^{-3} \sum_{\ell \in \mathbb{Z}^3} \left(\sum_{r,s,t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{ar^2(s^2 + t^2) + \ell_1 r + \ell_2 s + \ell_3 t}{q}\right) \right) \widehat{\Phi}_\beta\left(\frac{\ell}{q}\right).$$

Estimation of $\mathcal{B}_H(\theta)$: the major arcs

$$\mathcal{B}_H(\theta) = \frac{1}{2}q^{-3} \sum_{\ell \in \mathbb{Z}^3} \left(\sum_{r,s,t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{ar^2(s^2 + t^2) + \ell_1 r + \ell_2 s + \ell_3 t}{q}\right) \right) \widehat{\Phi}_\beta\left(\frac{\ell}{q}\right).$$

The contribution from $\ell \neq 0$ can be shown to be acceptable, while $\ell = 0$ gives

$$\underbrace{\frac{1}{2}q^{-3} \sum_{r,s,t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{ar^2(s^2 + t^2)}{q}\right) \widehat{\Phi}_\beta(0)}_{C_q(a)},$$

where

$$\widehat{\Phi}_\beta(0) = \int_{\mathbb{R}^3} \Phi_\beta(v, x, y) dv dx dy.$$

Estimation of $\mathcal{B}_H(\theta)$: the major arcs

Using

$$\int_{\mathbb{R}} e^{-zu^2} du = \sqrt{\frac{\pi}{z}}, \quad \operatorname{Re}(z) > 0,$$

we get

$$\begin{aligned} \frac{1}{2} \widehat{\Phi}_\beta(0) &= \frac{1}{2} \int_{\mathbb{R}^3} U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) \exp\left(-\frac{v^2}{H^2} + 2\pi i \beta v^2(x^2 + y^2)\right) dv dx dy \\ &= \frac{\sqrt{\pi}}{2} W(\beta H^4) H^3, \end{aligned}$$

where

$$W(t) := \int_{\mathbb{R}^2} U(x)U(y) (1 - 2\pi i t(x^2 + y^2))^{-1/2} dx dy.$$

Hence

$$\mathcal{B}_H(\theta) = \frac{\sqrt{\pi}}{2} C_q(a) W(\beta H^4) H^3 + \text{acceptable error}, \quad \forall \theta \in \mathfrak{M}.$$

Estimation of $\mathcal{B}_H(\theta)$: the major arcs

Recall

$$\mathcal{B}_H(\theta) = \frac{\sqrt{\pi}}{2} C_q(a) W(\beta H^4) H^3 + \text{acceptable error}, \quad \forall \theta \in \mathfrak{M},$$

with

$$\arg(W(t)) = \arg\left(\int_{\mathbb{R}^2} U(x)U(y)(1 - 2\pi it(x^2 + y^2))^{-1/2} dx dy\right) \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right].$$

If we can show $|\arg(C_q(a))| \leq \pi/4$, then $|\arg(\text{main term})| \leq \pi/2$ and thus

$$\operatorname{Re}(\mathcal{B}_H(\theta)) \geq \text{acceptable error}, \quad \forall \theta \in \mathfrak{M}.$$

$C_q(a)$ and the Gauss sum

Observe

$$C_q(a) = q^{-3} \sum_{r,s,t \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{ar^2(s^2 + t^2)}{q}\right) = q^{-3} \sum_{r \in \mathbb{Z}/q\mathbb{Z}} G_q(ar^2)^2,$$

where

$$G_q(a) := \sum_{x \in \mathbb{Z}/q\mathbb{Z}} e\left(\frac{ax^2}{q}\right)$$

is the quadratic Gauss sum.

Lemma 3.4 (Multiplicativity)

If $q = q_1 q_2$ with $(q_1, q_2) = 1$ and $(a, q) = 1$, then

$$C_q(a) = C_{q_1}(a\overline{q_2})C_{q_2}(a\overline{q_1}).$$

Facts about the Gauss sum

Suppose $a = p^t a_0$ with $t \geq 0$ and $p \nmid a_0$.

- $G_{p^s}(a) = p^t G_{p^{s-t}}(a_0)$ when $s > t$ and $G_{p^s}(a) = p^s$ when $0 \leq s \leq t$.
- $p > 2$:

$$G_{p^m}(a_0) = \begin{cases} p^{m/2}, & \text{if } m \text{ even,} \\ \left(\frac{a_0}{p}\right) \varepsilon_p p^{m/2}, & \text{if } m \text{ odd,} \end{cases}$$

where

$$\varepsilon_p = \begin{cases} 1, & \text{if } p \equiv 1 \pmod{4}, \\ i, & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

- $p = 2$:

$$G_{2^m}(a_0) = \begin{cases} 0, & \text{if } m = 1, \\ 2^{m/2}(1 + i^{a_0}), & \text{if } m \geq 2 \text{ is even,} \\ 2^{(m+1)/2}e(a_0/8), & \text{if } m \geq 3 \text{ is odd.} \end{cases}$$

$$|\arg(C_q(a))| < \pi/4$$

When $p \nmid a$, the terms in

$$C_{p^s}(a) = p^{-3s} \sum_{r \in \mathbb{Z}/p^s\mathbb{Z}} G_{p^s}(ar^2)^2$$

with $p^s \mid r^2$ contribute $p^{-3s} \cdot p^{s - \lceil s/2 \rceil} \cdot p^{2s} = p^{-\lceil s/2 \rceil}$.

- $p > 2$: $C_{p^s}(a) \geq p^{-2\lceil s/2 \rceil} > 0$.
- $p = 2$: $\operatorname{Re}(C_{2^s}(a)) = 2^{-\lceil s/2 \rceil}$, $\operatorname{Im}(C_{2^s}(a)) = 2^{-\lceil s/2 \rceil} - 2^{-s}$, and so

$$|\arg(C_{2^s}(a))| < \frac{\pi}{4}.$$

Combined with multiplicativity, this gives $|\arg(C_q(a))| < \pi/4$.

Estimation of $\mathcal{B}_H(\theta)$: the minor arcs

Given $H \geq 1$, and $0 < \sigma < 1/4$, define the minor arcs $\mathfrak{m} := \mathbb{T} \setminus \mathfrak{M}$, where

$$\mathfrak{M} = \bigcup_{1 \leq q \leq Q} \bigcup_{\substack{1 \leq a \leq q \\ (a,q)=1}} \left\{ \theta \in \mathbb{T} : \left\| \theta - \frac{a}{q} \right\| \leq \frac{Q}{qH^4} \right\}, \quad Q = H^{1-4\sigma}.$$

For each $\theta \in \mathfrak{m}$, Dirichlet's theorem produces a reduced fraction a/q such that

$$Q < q \leq \frac{H^4}{Q}, \quad \left| \theta - \frac{a}{q} \right| \leq \frac{Q}{qH^4} \leq \frac{1}{q^2}.$$

With $V = H(40 \log H)^{1/2}$, we have

$$\mathcal{B}_H(\theta) \approx E(\theta) := \frac{1}{2} \sum_{x,y \in \mathbb{Z}} U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right) \sum_{|v| \leq V} e^{-(v/H)^2} e(v^2(x^2 + y^2)\theta).$$

Estimation of $\mathcal{B}_H(\theta)$: the minor arcs

$$E(\theta) = \frac{1}{2} \sum_{x,y \in \mathbb{Z}} \overbrace{U\left(\frac{x}{H}\right) U\left(\frac{y}{H}\right)}^{\text{bounded and supported on } [H, 2H]} \sum_{|v| \leq V} e^{-(v/H)^2} e(v^2(x^2 + y^2)\theta).$$

By Cauchy–Schwarz,

$$\begin{aligned} |E(\theta)|^2 &\ll_U H^2 \sum_{H \leq x, y \leq 2H} \left| \sum_{|v| \leq V} e^{-(v/H)^2} e(v^2(x^2 + y^2)\theta) \right|^2 \\ &= H^2 \sum_{|v_1|, |v_2| \leq V} \overbrace{e^{-(v_1/H)^2} e^{-(v_2/H)^2}}^{\leq 1} \sum_{H \leq x, y \leq 2H} e((v_1^2 - v_2^2)(x^2 + y^2)\theta) \\ &\leq H^2 \sum_{|n| \leq V^2} \sum_{\substack{|v_1|, |v_2| \leq V \\ v_1^2 - v_2^2 = n}} \left| \sum_{H \leq x, y \leq 2H} e(n(x^2 + y^2)\theta) \right|. \end{aligned}$$

Estimation of $\mathcal{B}_H(\theta)$: the minor arcs

$$\begin{aligned}
 |E(\theta)|^2 &\ll_U H^2 \sum_{|n| \leq V^2} \sum_{\substack{|v_1|, |v_2| \leq V \\ v_1^2 - v_2^2 = n}} \left| \sum_{H \leq x, y \leq 2H} e(n(x^2 + y^2)\theta) \right| \\
 &= H^2 \sum_{1 \leq |n| \leq V^2} \sum_{\substack{|v_1|, |v_2| \leq V \\ v_1^2 - v_2^2 = n}} \left| \sum_{H \leq x \leq 2H} e(nx^2\theta) \right|^2 + \underbrace{O(VH^2)}_{\text{contribution from } n=0} \\
 &\ll V^{o(1)} H^2 \sum_{1 \leq n \leq V^2} \left| \sum_{H \leq x \leq 2H} e(nx^2\theta) \right|^2 + VH^2,
 \end{aligned}$$

which can be estimated in a standard way.

Thank you!